

Population diversity in Pacific herring of the Puget Sound, USA

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Received: 20 November 2014 / Accepted: 30 August 2015 / Published online: 1 October 2015
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Abstract Demographic, functional, or habitat diversity can confer stability on populations via portfolio effects (PEs) that integrate across multiple ecological responses and buffer against environmental impacts. The prevalence of these PEs in aquatic organisms is as yet unknown, and can be difficult to quantify; however, understanding mechanisms that stabilize populations in the face of environmental change is a key concern in ecology. Here, we examine PEs in Pacific herring (*Clupea pallasii*) in Puget Sound (USA) using a 40-year time series of biomass data for 19 distinct spawning population units collected using two survey types. Multivariate auto-regressive state-space models show independent dynamics among spawning subpopulations, suggesting that variation in herring production is partially driven by local effects at spawning grounds or during the earliest life history stages. This independence at the subpopulation level confers a stabilizing effect on the overall Puget Sound spawning stock, with herring being as much as three times more stable in the face of environmental perturbation than a single population unit of the same size. Herring populations within Puget Sound are highly asynchronous but share a common negative growth rate

and may be influenced by the Pacific Decadal Oscillation. The biocomplexity in the herring stock shown here demonstrates that preserving spatial and demographic diversity can increase the stability of this herring population and its availability as a resource for consumers.

Keywords Forage fish · *Clupea pallasii* · Population structure · State-space model · Time series analysis · Portfolio effect

Introduction

A prevailing concept in ecology is the positive relationship between biodiversity and stability (MacArthur 1955; Hooper et al. 2005). Biodiversity can manifest as genetic, habitat, life history, phenotypic, or species diversity. At a community level, species-rich communities are thought to be more stable because functionally similar species can have complementary dynamics (Lehman and Tilman 2000). At a population level, genetic heterogeneity within species can dampen the variance in abundance and prevent extinction (Agashe 2009). Moreover, asynchrony, weak synchrony, or independence among individual population components can cause population stability at landscape scales. Metapopulation dynamics can increase overall population stability by providing immigrant individuals to declining or “sink” patches (Hanski 1999; McQuinn 1997). Distinct habitats that filter regional environmental processes, variable responses to environmental conditions, and population stochasticity can also produce diverse population dynamics (Schindler et al. 2015). Across multiple mechanisms and scales, species and population components may fluctuate independently or asynchronously from each other, thereby dampening the aggregate community or population

Communicated by Brian Shuter.

Electronic supplementary material The online version of this article (doi:10.1007/s00442-015-3439-7) contains supplementary material, which is available to authorized users.

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response to perturbation. These stabilizing dynamics are referred to as “portfolio effects” (PEs) (Schindler et al. 2010) and can serve as an organizing concept for understanding how organisms might fare in the face of future environmental change (Schindler et al. 2015).

PEs may be important for the reliability of spatially or temporally structured populations, and for the stability of food webs where predators exploit prey that are spatially or temporally organized (Schindler et al. 2013). Pacific herring (*Clupea pallasii*), a major forage fish species in the eastern Pacific Ocean, are food for predators across their range (around the Pacific Rim from Southern California to Korea), including marine mammals (Trites et al. 2007; Lance and Jeffries 2006), seabirds (Hatch 2013; Schrimpf et al. 2012), and fish (Beaudreau and Essington 2011; Duffy et al. 2010). During the months when herring spawn in the shallow subtidal zone, their eggs are also a food source for seabirds (Anderson et al. 2009). Pacific herring support valuable commercial fisheries: landings of Pacific herring in the US alone were worth \$17 million in 2013 (National Marine Fisheries Service 2013). Because of their ecological and economic importance, and because their life history leaves them highly vulnerable to human activities (Shelton et al. 2014), maintaining the reliability of the Pacific herring resource is a high priority in marine ecosystems where they occur.

Biocomplexity [defined here as properties emerging from complex behavioral, biological and physical interactions (Michener et al. 2001)] in marine fishes is still not widely understood, though marine fishes often exhibit subpopulation spatial and demographic diversity. While fine-scale genetic diversity has been demonstrated in other marine fishes [e.g., North Atlantic cod (Hutchinson et al. 2001)], recent studies have also revealed that source-sink dynamics can occur within herring populations that are genetically homogeneous (Lindgren et al. 2014). These dynamics are relevant to management decisions: population diversity can occur within groups that are managed as a single stock (Casini et al. 2011), and ignoring this complexity can result in the overfishing of sensitive subpopulations and slow recovery of overfished stocks (Hutchinson 2008). Because adult herring are highly mobile, there is little reason to suspect that there is stock structure in regional population dynamics. Conversely, because there appears to be some fidelity to spawning areas (Hay et al. 2001), it is possible that differential geographic site selection and spawn timing is strong enough to maintain subpopulation diversity (Grabowski et al. 2011). Herring populations display differences in spawning timing and behavior, producing a situation where PEs have the potential to stabilize regional abundances.

The Puget Sound, a network of coastal waters which spans from northwestern Washington State, USA, to the southwestern tip of British Columbia, Canada, is an

urbanizing estuary containing at least 21 geographically distinct spawning groups of Pacific herring (Fig. 1); here we refer to these spatially distinct spawning groups as “subpopulations,” in the sense that they are components of the Puget Sound population. We refer to hypothesized groupings of subpopulations being tested for shared population structure as “population units.” These hypothesized population units are based on the work of other scholars who have defined population units according to different shared characteristics, such as microsatellite markers, muscle tissue contaminants, and geography. Puget Sound herring display several characteristics of stock structure, including differences in weight and length at age, spawn location and timing, and pre-spawning behavior (Chapman et al. 1941). However, the consequences of this diversity for the stability of the overall Puget Sound herring resource have not been quantified.

Here, we use a time series method for describing population structure, a multivariate auto-regressive (AR) state-space (MARSS) model (Hinrichsen and Holmes 2009), incorporating data from two survey methods (described below), accounting for autocorrelation in annual abundance data and variability in error rates between the two survey methods. Using 40 years of Puget Sound herring biomass data, we address the following questions:

1. What is the population structure of Puget Sound Pacific herring? How many population units exist within Puget Sound?
2. Does the population structure of Puget Sound herring confer a stabilizing PE on the population?

Last, because previous work has shown the influence of regional environmental drivers on Puget Sound herring recruitment (Reum et al. 2011), we also include regional environmental data in our model to ask:

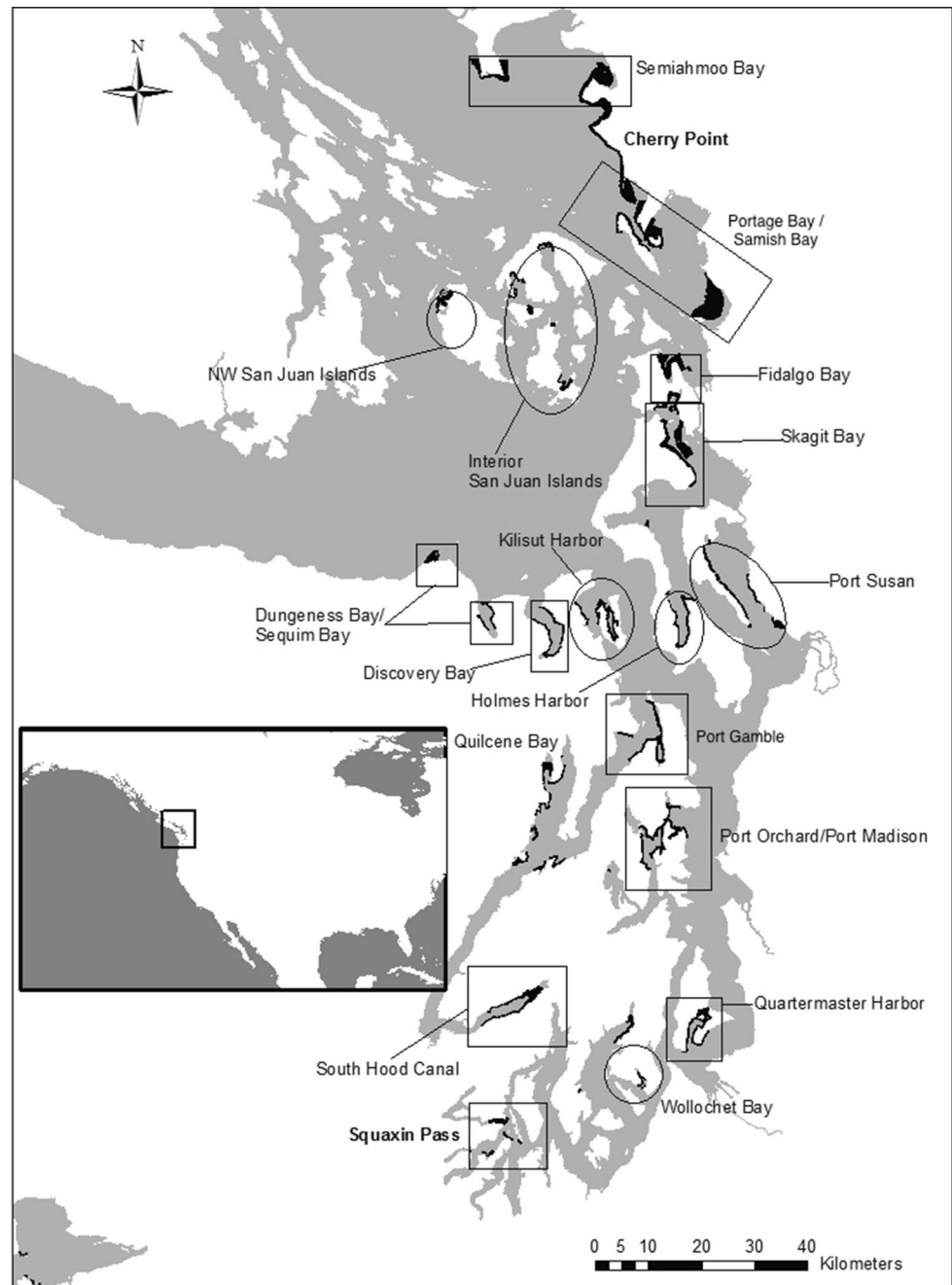
3. Are there regional environmental conditions that explain variation in abundance across subpopulations?

Materials and methods

Data collection

We studied population structure in Puget Sound Pacific herring using time series data of adult spawner biomass from historical surveys conducted by the Washington Department of Fish and Wildlife (WDFW) from 1973 to 2012 on 21 Pacific herring spawning grounds (Fig. 1). Two methods have been used historically to estimate spawning biomass: spawn deposition surveys and acoustic/trawl

Fig. 1 Pacific herring (*Clupea pallasii*) spawning sites in Puget Sound (black shaded areas). Cherry Point, Semiahmoo, and Squaxin Pass are often designated as separate subpopulations from the rest of the Puget Sound populations



surveys (Stick et al. 2014). Spawn deposition surveys are conducted at each spawning site throughout the spawning season. Surveyors drag a rake attached to a line along the sea floor in subtidal to lower intertidal zones, and visually, qualitatively estimate the density of eggs attached to marine vegetation or cobble. Qualitative egg density is converted to a quantitative estimate using standard conversion factors and then extrapolated to egg abundance for the full spawning site using the length of the shoreline covered by spawn and a constant width of egg deposition (Stick et al. 2014). Beaches are surveyed once or twice weekly during

the spawning season to estimate cumulative spawn deposition. Adult spawner biomass for the site is then calculated as total egg abundance divided by a constant fecundity (eggs/g adult body mass).

Spawning adults were also historically surveyed (until 2009) using ship-based acoustics in pre-spawning holding areas, where herring aggregate 1–2 months prior to spawning. Acoustic measurements of abundance from backscatter were converted to herring biomass based on size information from paired trawl catches. For the present study, we limited our analyses to data representing only sites: (1) in

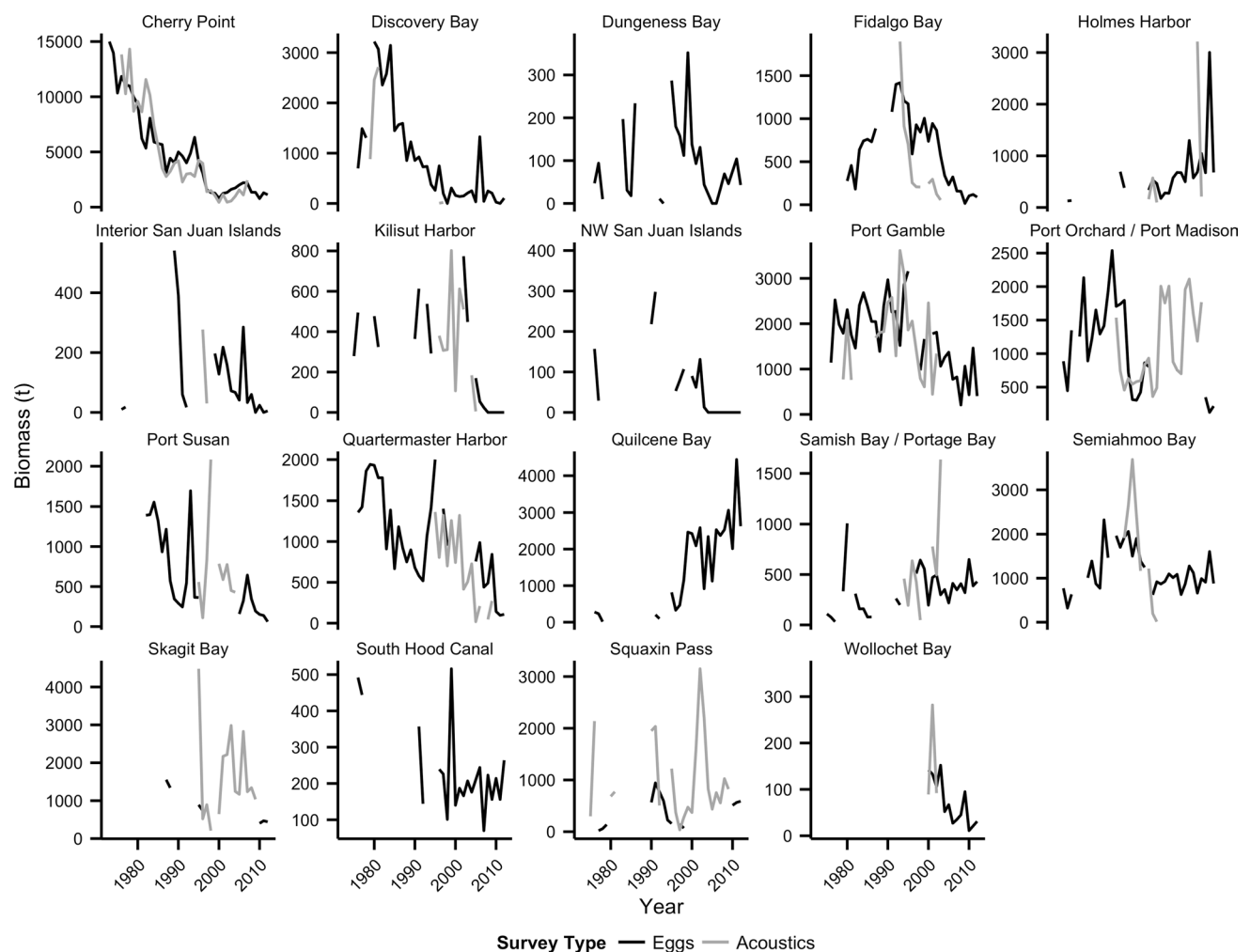


Fig. 2 Adult herring spawner biomass estimates from egg deposition surveys (black lines) and acoustic/trawl surveys (grey lines) in Washington State (data are from Washington Department of Fish and Wildlife 2013). Only the 19 population units used in the analysis are shown here

Puget Sound, and (2) with >10 years of historical data. This resulted in a data set of 19 sites; hereafter reference will be made only to these 19 subpopulations.

Previous efforts to describe Puget Sound herring population trends and structure have been hampered by the fact that herring are surveyed using two different methods. The two methods estimate abundance differently over space and time (Fig. 2): egg deposition surveys take place during the spawning season, while acoustic surveys of pre-spawning aggregations occur before the onset of spawning at each site and less frequently. It is possible that egg surveys and acoustic surveys capture different processes: non-spawning herring, or spawning herring ultimately destined for other spawning grounds, may also aggregate at pre-spawning aggregation areas, whereas egg surveys are less likely to overestimate spawning biomass. Errors in acoustic surveys are likely sensitive to the timing of surveys relative to the peak of spawning activity (Schweigert and Haegele

2001), and egg surveys could be biased by estimates of total spawning area, fecundity (Hay 2011), and skewed sex ratio. Previous comparisons of the data from both methods have indicated substantial differences in biomass estimates between acoustic and egg surveys (Burton 1990).

Because previous research has shown that early life stages of herring are correlated with regional climate drivers (Reum et al. 2011), we included the intensity of coastal upwelling (NOAA Fisheries ERD environmental data; available upon request from <http://oceanview.pfeg.noaa.gov/erddap/search/index.html?searchFor=upwelling+index>) and the Pacific Decadal Oscillation index (PDO; see <http://jisao.washington.edu/pdo/PDO.latest>) in the model as regional covariates of herring biomass. Since each index provided monthly data, indices were averaged over winter spawning months (January–March) to obtain an annual winter mean. Regional covariate data (winter PDO and upwelling during rearing) were available for the entire time

series, and we assumed they affected all 19 subpopulations. We applied lags to the regional covariates from 2 years (the minimum number of years to maturity for Pacific herring) to 7 years (the oldest age class of herring in these subpopulations, on average) to assess the impact of juvenile growth conditions on adult spawning biomass.

Modeling approach

We used the MARSS framework (Hinrichsen and Holmes 2009) to model stock structure and estimate population growth parameters in Puget Sound herring, while accounting for process variability and observation error. The MARSS framework is based on the Gompertz population growth model ($x_t = bx_{t-1} + u$, in which x_t is the log of the population size at time t , b is inversely proportional to the magnitude of density dependence, and u is the rate of exponential growth or decline). The Gompertz model describes a population experiencing density-dependent growth with stochastic population dynamics (Ives et al. 2003). The population is driven both by environmental variation and demographic variation. The Gompertz model can be elaborated to describe a multi-stock system, and the model implies that the natural log of population size can be described by an AR(1) process (Holmes et al. 2012). The model consists of a univariate state-space model that incorporates stochasticity in population growth (process error), as well as variation in observations due to subsampling from the whole population, differences in survey methods (i.e., acoustic survey vs. egg survey), and inaccurate counting during either survey method (observation error). The univariate state-space model is then modified to accommodate count data from multiple sites, where sites may or may not represent independent subpopulations (Hinrichsen and Holmes 2009). The basic form of the model includes a process equation that describes the AR process, as

$$\mathbf{x}_t = \mathbf{B}\mathbf{x}_{t-1} + \mathbf{u} + \mathbf{C}\mathbf{c}_t + \mathbf{w}_t \quad \mathbf{w}_t \sim \text{MVN}(0, \mathbf{Q}) \quad (1)$$

and an observation equation,

$$\mathbf{y}_t = \mathbf{Z}\mathbf{x}_t + \mathbf{a} + \mathbf{v}_t \quad \mathbf{v}_t \sim \text{MVN}(0, \mathbf{R}) \quad (2)$$

which describes observations of the process. In the process model (Eq. 1), $\mathbf{x}_t$ is an $m \times 1$ vector of the natural log of biomass for each of the m population units at time t (the values of x_t are referred to as “states”). $\mathbf{B}$ is an $m \times m$ matrix whose elements are coefficients describing the degree to which each population unit reverts to a long-term mean, $\mathbf{u}$ is an $m \times 1$ vector of population unit growth rates, $\mathbf{C}$ is an $m \times p$ matrix of the effects of p covariates on herring biomass and $\mathbf{c}_t$ represents covariate observations at time t . Process error (the vector $\mathbf{w}_t$) is assumed to be uncorrelated in time and drawn randomly from a multivariate normal distribution with mean zero and variance–covariance

matrix $\mathbf{Q}$. Because the Gompertz process is linear in log space, Gaussian errors are appropriate (Ives et al. 2003). To confirm that stationarity was an adequate assumption for our models, we used augmented Dickey-Fuller tests on the residuals from the states of the best-fit model [Electronic supplementary material (ESM) Table S1]. Only three of the 19 subpopulations (Kilisut Harbor, Northwest San Juan Islands, and Wollochet Bay) had non-stationary residuals from the states, so we assumed that stationarity was an adequate assumption for our models.

In the observation model (Eq. 2), population data from the n subpopulations enter the model as $\mathbf{y}_t$, an $n \times 1$ vector of observed log adult spawning herring biomass for each of $n = 19$ discrete WDFW-surveyed herring subpopulations in the Puget Sound at time t . $\mathbf{Z}$ is an $n \times m$ matrix of 0 s and 1 s that describes which observation time series are associated with each population unit, $\mathbf{a}$ is an $n \times 1$ vector representing the bias between the observations and the process, and observation error ($\mathbf{v}_t$) is temporally uncorrelated and drawn from a multivariate normal distribution, with variance–covariance matrix $\mathbf{R}$.

We included the influence of environmental covariates in a subset of the models tested to determine whether environmental data improved our model fit. Matrix $\mathbf{C}$ (Eq. 1) indicates the effect of each covariate on each state (x_t). We assumed that covariate data were observed without error, so covariate states are described in the process equation as $\mathbf{c}_t$. This model was used to test hypotheses about stock structure, survey error, and response to environmental factors. We used the expectation–maximization (EM) algorithm to obtain maximum likelihood estimates of parameters (Holmes 2014). In the EM algorithm we used 2000 iterations.

Stock structure

To determine stock structure and estimate abundance trends of Puget Sound herring, we used data from egg deposition surveys, because they were the most complete data, and fit models modified to represent different hypotheses of stock structure (Table 1; ESM Table S2; ESM Appendix S3). The basic model used to detect stock structure is given by Eqs. 1 and 2 without the covariate terms: $\mathbf{a}$ is treated as a scaling parameter (the first element sets the bias to zero, and other elements are estimated relative to the first) and $\mathbf{B}$ is an identity matrix, i.e., no interactions among subpopulations. The $\mathbf{Z}$ matrix and $\mathbf{u}$ vector were then modified to test hypotheses about stock structure, by changing the number of m population units according to each hypothetical population structure. Different values of m correspond to different numbers of columns in the $\mathbf{Z}$ matrix; when the populations are modeled as independent, $m = n$ and $\mathbf{Z}$ is an $n \times n$ identity matrix; when there are $m < n$ hypothesized

Table 1 Structure and ranking of multivariate autoregressive state-space models fitted to Pacific herring biomass time series data in the Puget Sound

Model	Hypothesized population structure	Number of population units (m)	Covariance ($\mathbf{Q}_{\text{offdiag}}$)	Δ_{AICc}	Δ_{AICb}	Relative likelihood
1	Each subpopulation separate (no covariance)	19	0	2.6	0.0	1
2	Each subpopulation separate (covariance is equal)	19	Equal	0.0	71.6	2.82 E-16
3	Basins	5	0	185.7	178.8	4.69 E-41
4	All one population (panmictic)	1	0	198.0	365.0	9.65 E-44
5	Microsatellites (Cherry Point, Squaxin Pass, and all others)	3	0	168.8	378.7	2.20 E-37
6	Regions (North Puget Sound, South/Central Puget Sound, Straits)	3	0	199.2	393.8	5.28 E-44
7	Contaminants (Cherry Point, Semiahmoo, all others)	3	0	181.5	446.4	3.75 E-40
8	Each subpopulation separate with different trends (different trends, no covariance)	19	0	20.9	1363.8	2.82 E-05
9	Each subpopulation separate with different trends, different variance and covariance for each subpopulation	19	All different (all elements of $\mathbf{Q}$ estimated)	515.7	28468.9	0.000

For all models, process variance and observation error are equal across sites. Observation error is not correlated between sites ($\mathbf{R}_{\text{offdiag}} = 0$). The first column indicates the way observation time series were assigned to process time series. For more detailed descriptions of model structure, see Electronic supplementary material, Appendix S3

AIC Akaike information criterion

population units, $\mathbf{Z}$ has m columns. For each model, a different $\mathbf{Q}$ ($m \times m$) and u vector ($m \times 1$) were estimated.

Although there are several possible hypothesized population structures, we chose to evaluate a subset of nine models based on the best available information. First, we tested two “null” models; one in which all subpopulations fluctuate independently (i.e., each subpopulation follows a different process; model 1), and one in which all subpopulations represent samples from a single underlying process (model 4). In the two null models, 19 separate populations were modeled as $m = 19$ (model 1), and one panmictic population was modeled as $m = 1$ (model 4). We also included a model where populations were independent but covariance was equal (model 2) and one in which populations were independent but covariance was 0 (model 8).

We also tested four models of hypothesized population structure previously suggested based on genetic data (Small et al. 2005; model 5), contaminant signatures (West et al. 2008; model 7), and spatial structure (models 3 and 6; Stout et al. 2001; Penttilä 2007). The genetic structure hypothesis, based on microsatellite loci, suggests three genetic units ($m = 3$): distinct units in north Puget Sound (Cherry Point) and the south Puget Sound (Squaxin Pass), and all other subpopulations aggregated into one unit (Small et al.

2005). The contaminant hypothesis states that Puget Sound herring absorb contaminants at their feeding grounds, dividing subpopulations into three ecological units ($m = 3$) (West et al. 2008). West et al. (2008) found elevated levels of polychlorinated biphenyls (PCBs) and dichlorodiphenyl-trichloroethane in Puget Sound herring subpopulations south of the Cherry Point stock (in bold; Fig. 1). Semiahmoo has lower concentrations of PCBs than the other subpopulations, and is hypothesized to be a separate population unit. The third hypothesis spatially separates all herring subpopulations into three regional units ($m = 3$; Penttilä 2007): North Puget Sound, South/Central Puget Sound, and Strait of Georgia. The fourth hypothesis we tested aggregates the subpopulations into the five Puget Sound basins defined by NOAA: North Puget Sound, Whidbey, Main, Hood Canal, and South Puget Sound ($m = 5$) (model 3; Stout et al. 2001). The subpopulations belonging in each grouping are given in ESM Table S2. Finally, we expected that covariance might differ between pairs of stocks, so we also fit a model that estimated all elements of $\mathbf{Q}$ (model 9).

For each of the simpler hypothesized population structures (model 1 and models 3–8), we assumed no covariance between groups. We also compared these population structure models under conditions where covariance was allowed between

groups, to make sure that assumptions about the amount of covariance were not impacting our ability to find the structure with the best-supported number of population units. Because only the egg survey data were used to compare the different population structure hypotheses, we assumed that observation errors (the diagonal elements of **R**) were equal across sites. For all but one model, we estimated one parameter for the process variance (the diagonal of the variance–covariance matrix; **Q**_{diag}) and one parameter for covariance between stocks (the off-diagonal of **Q**; **Q**_{offdiag}). We examined residuals from the best-fit models to confirm normality and determine stationarity.

Portfolio effects

PEs in ecology can be calculated simply by comparing the temporal coefficient of variation (CV) of an aggregate population with the average CV of all the subpopulations (e.g., Schindler et al. 2010). However, CV is often size dependent in ecological systems: as the abundance of a single subpopulation grows, its SD usually increases non-linearly according to Taylor’s power law (McArdle et al. 1990), so larger populations will be less variable than expected using the average-CV PE. A second method for calculating PEs accounts for this by extrapolating the mean–variance relationship to the aggregate population size, which can then be compared to the observed variability of the aggregate population. The ratio of observed to predicted variance is the mean–variance PE (Anderson et al. 2013). The mean–variance PE is calculated using the following linear regression:

$$\ln(\sigma_i^2) = \beta_0 + z \cdot \ln(\mu_i) + \varepsilon_i \quad (3)$$

where $\ln(\mu_i)$ and $\ln(\sigma_i^2)$ are the interannual log mean and log variance, respectively, of the subpopulation time series (*i*) from the state-space model, *z* is the slope of the linear regression between $\ln(\mu_i)$ and $\ln(\sigma_i^2)$ and ε is independent and identically distributed residual error with mean 0 and an estimated variance. Equation 3 is used to predict the overall variance $\hat{\sigma}^2$ expected from a homogeneous population of the aggregated subpopulations, and the ratio of observed σ^2 to predicted $\hat{\sigma}^2$ is the mean–variance PE. PEs can only be quantified when the population size is known at each time step, which is rare in long-term ecological time series. Therefore, we used MARSS model fits from the best-fit stock structure model for each population unit as the basis for PE calculation.

We used Spearman’s rank correlation to determine the strength of covariance between subpopulations, and evaluated community-wide synchrony using the index proposed by Loreau and de Mazancourt (2008):

$$\varphi_N = \frac{\sigma_{NT}^2}{(\sum_i \sigma_{N_i})^2} \quad (4)$$

where σ_{NT}^2 is the aggregate variance of the total biomass across all subpopulations and σ_{N_i} is the SD of the biomass of each subpopulation *i*. This “synchrony index” varies between 0 (perfect asynchrony) and 1 (perfect synchrony). Average-CV and mean–variance PE, and synchrony index were calculated using the *ecofolio* package in R (Anderson et al. 2013).

Comparison of survey methods

To compare the effectiveness of each herring survey method for estimating adult spawner biomass, we modeled the following potential scenarios:

1. Egg and acoustic surveys measure the same process, and observation error for each survey type does not vary by site.
2. Egg and acoustic surveys measure different processes, and observation error for each survey type does not vary by site.
3. Egg and acoustic surveys measure the same process, and observation error magnitude varies by site.
4. Egg and acoustic surveys measure different processes, and observation error magnitude varies by site.

The structure of the **Q**, **R**, **Z**, and **B** matrices were modified to test hypotheses about the process (or processes) captured by each survey type, and the error of each survey type in measuring these. Model structures tested here were informed by the best-fit stock structure model: beaches were assumed to be independent and have similar process variances. We used different structures of **Q**, **R**, **Z**, and **B** to test the error associated with each survey type in estimating the underlying state process for each population unit. When comparing egg and acoustic/trawl survey data, we assumed that population growth trends (*u*) and process variance (**Q**_{diag}) were equal, and that there was no process error covariance (**Q**_{offdiag} = 0).

Environmental effects on herring abundance

To test for environmental drivers of herring abundance trends, beginning with the best-fit stock structure model from the “[Stock structure](#)” section (19 independent population units; referred to here as the null model), we added environmental data (the “full model”; data descriptions below) to see if they improved model fits of biomass time series data: regional environmental conditions (PDO and/or upwelling). If covariates explained variation in biomass, we considered the full model a better fit to the biomass data than the null model.

For regional covariates, the null model (herring biomass only) was compared to models that included PDO,

upwelling, and PDO plus upwelling. The spawning biomass in any given year is composed of individuals aged 2 years and older; therefore the impact of regional environmental factors on spawning biomass includes impacts on eggs and larvae on spawning grounds, and impacts on adult fish in the open ocean during that year.

Model selection

Akaike's information criterion (AIC) is often used to compare the difference in likelihood between an estimated model relative to the distribution of possible data from a true (unknown) process. AIC_c can underestimate the complexity penalty for MARSS models when data sets are small, favoring more complex models (Ward et al. 2010). AIC calculations include a likelihood term and a penalty for model complexity, which can be estimated using bootstrapping (AIC_b) (Cavanaugh and Shumway 1997) or Monte Carlo simulation. In order to properly penalize for the number of parameters estimated and compare models with different **Z** matrices, we used a bootstrap AIC (AIC_b) (Hinrichsen and Holmes 2009), which is designed for state-space models and has been used in previous studies to compare state-space models with different **Z** matrices (e.g., Hinrichsen and Holmes 2009; Ward et al. 2010). This calculation uses bootstrapping to estimate the penalty term for the number of parameters used. Models with different **Z** matrices do not contain different data; **Z** is a design matrix that specifies the relationship between the biomass time series and the hypothesized subpopulations to which they belong.

Results

Synchrony, stock structure and PEs

MARSS model fits indicated that there are 19 population units of herring within Puget Sound, which behave independently (Table 1; Fig. 3). The best-fit model for herring stock structure described Puget Sound herring as consisting of 19 independent population units (i.e., each subpopulation was one population unit, **Z** was identity and $m = 19$), with an overall negative growth rate u [$u = -0.01$; 95 % confidence interval $(-0.03, 0.01)$]. Genetic, contaminant, regional, and NOAA basin population unit designations did not fit biomass time series as closely, based on AIC_b scores (Table 1). Process variability, which was also shared by different population units, was very large compared to the negative deterministic trend: process variance $\mathbf{Q}_{\text{diag}} = 0.09$ (0.04, 0.10) ($\text{sd} = 0.3$). The preferred model estimated no covariance

across all units ($\mathbf{Q}_{\text{offdiag}} = 0$), meaning that while the population units fluctuate at similar magnitudes each year (based on the $\mathbf{Q}_{\text{diag}}$), they do not covary. In this analysis, because the covariance $\mathbf{Q}_{\text{offdiag}}$ is set to the same value across all pairs of population units, it is impossible to tell whether covariance is low across all population units or whether there is covariance among some population units and not others. A model that estimated different deterministic trends (u) for each population unit did not improve the model fit (model 8; Table 1). A model that estimated individual variance and all pairwise covariances (model 9; Table 1) did not improve the model fit (variance–covariance matrix in ESM Table S3). Observation error for egg survey estimates of biomass was $\mathbf{R}_{\text{diag}} = 0.25$ (0.21, 0.31). Residuals from the best-fit model appeared normal and uncorrelated.

Synchrony between population units was relatively low (Loreau and de Mazancourt synchrony index = 0.23, where 1 is a completely synchronous population and 0 is completely asynchronous; Fig. 4). Correlations between population units were highly variable and were not dependent on geographical distance (Fig. 4; Mantel test $p = 0.69$). Negative correlations were strongest between population units that were “recovering” (i.e., biomass was increasing; Holmes Harbor, Quilcene Bay, Samish Bay/Portage Bay, and Squaxin Pass) and all other population units. The mean–variance PE calculated for Puget Sound herring was 1.92 (1.06, 3.45), meaning that the aggregate population ($\text{CV} = 0.30$) is 1.06–3.45 times more stable than one would expect from a homogeneous population with the same total biomass as the Puget Sound-wide herring metapopulation (expected $\text{CV} = 0.59$; Fig. 5).

Comparison of survey methods

There was strong support for the hypothesis that eggs and acoustic surveys measure the same process. In comparisons of egg deposition surveys versus acoustic surveys, the best-supported model ($\Delta_{\text{AICb}} = 0$) included identical underlying processes measured by both egg and acoustic surveys (Table 2). There was little support for the hypothesis that eggs and acoustic surveys measure different processes, even when observation error was allowed to vary by spawning site. In the best-fit model, the observation error from egg surveys ($r_{\text{egg avg.}} = 0.31$) was lower than the observation error from acoustic surveys ($r_{\text{acoustics avg.}} = 0.82$).

There was also some support for the hypothesis that the two survey types vary in their accuracy depending on the beach. Survey error rates varied by beach and method; measurement error was consistently higher for biomass

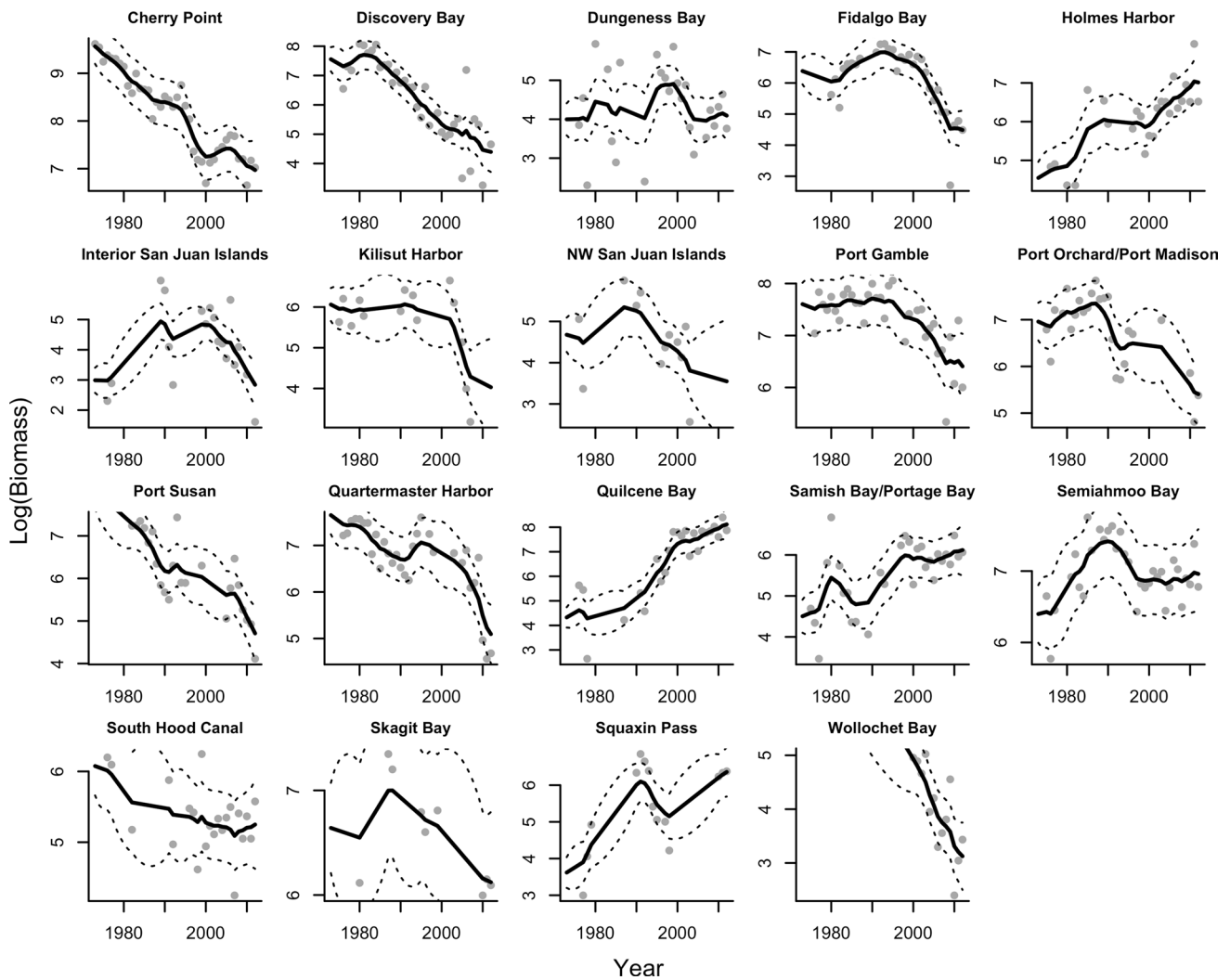


Fig. 3 Model fits (black lines) $\pm 95\%$ confidence intervals (CIs dashed lines) for the best-fit stock structure model (model 1; Table 1) of adult herring spawner biomass, based on egg survey data (grey points). Only the 19 population units used in the analysis are shown here

estimates from acoustic surveys, but was significantly higher at Semiahmoo Bay, Skagit Bay, and Quartermaster Harbor (Fig. 6).

Effects of regional environmental covariates

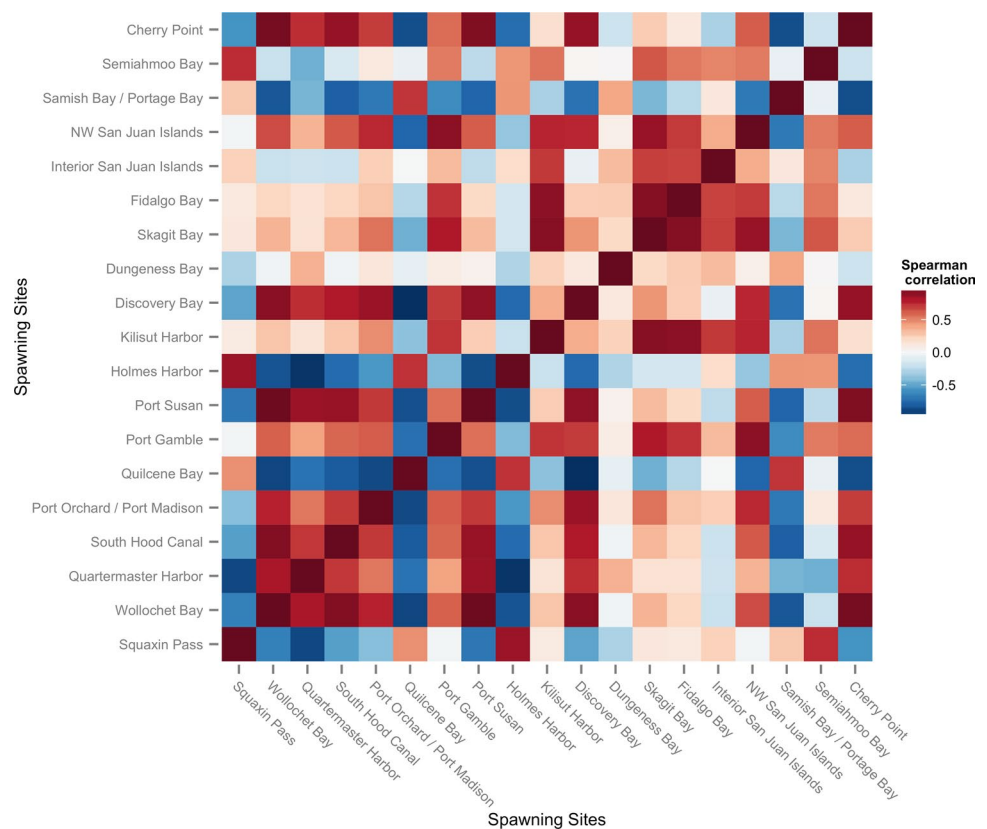
The inclusion of PDO as a covariate marginally improved model fits of population dynamics. The best-fit model included PDO as a covariate ($\Delta_{AICb} = 0$), with an estimated coefficient of -0.0024 ($-0.0042, -0.0006$). Including both PDO and upwelling as covariates did not improve model fits ($\Delta_{AICb} = 2.94$). This relationship was largely the same when different lags were applied to PDO and upwelling: regardless of the lag used to relate biomass to

covariates, herring-only models or models including PDO as a covariate were always best fit to biomass dynamics.

Discussion

Spatially distinct population dynamics can be stabilizing for fish species, but they are difficult to measure in pelagic stocks. Multivariate state-space models provide an opportunity to use data from diverse sources to elucidate the underlying population dynamics of a dominant forage fish. Our analyses show that spatial and temporal diversity in subpopulations of Puget Sound herring produce stability in the overall population. We found a common basin-wide

Fig. 4 Spearman's rank correlations among Puget Sound herring population units between 1973 and 2012, based on the best-fit stock structure model (model 1; Table 1). Population units are ordered from south (bottom and left) to north (top and right). The color of each square reflects the correlation coefficient (−1 to 1) calculated between a row population and a column population



trend in growth, with independence among subpopulations at a fine spatial scale, suggesting that drivers at multiple scales influence Puget Sound herring dynamics. This matches what is known about their life history, which includes life stages that are tightly linked to both shorelines and pelagic environments. We show here that though herring subpopulations have similar growth rates and variability, they exhibit independent dynamics within Puget Sound. This independence could result from multiple, unknown, local processes, differences in local responses to broader scale processes, movement, or stochasticity. This complexity is ultimately an advantage, because according to our analysis, it confers a degree of stability on the Puget Sound herring resource. Our results suggest that common oceanographic processes are not strong drivers of herring biomass. Furthermore, PEs may be an effective way to describe emergent dynamics in the absence of a good mechanistic population model.

PEs, asynchrony and stock structure

The independence we measured among Puget Sound spawning subpopulations has a stabilizing effect on the total Puget Sound herring population. The portfolio analysis indicated that the aggregate population

is 1.1–3.5 times more stable than a single population of the same size (Fig. 5). PEs can arise from a variety of mechanisms, including stochasticity, interactions among populations as in a metapopulation, and habitat-mediated or population-level variability in responses to environmental drivers (Schindler et al. 2015). Whether the PE observed here is due to genetic or environmental differences is unknown, but may be elucidated by examining differences in demographic rates between population units (Thorson et al. 2014). Regardless of the source of differences among subpopulations, our results suggest that it is important to preserve multiple spawning sites to ensure stable dynamics for Puget Sound herring and the ecosystem services they provide. Particularly for forage fish species, which demonstrate wide fluctuations in abundance, retaining some stability may be critically important for these populations' long-term prospects. Furthermore, retaining the availability of herring as a resource across a broad geographic range ensures that local drivers of decline or extinction do not cascade throughout a food web containing mobile predators that rely upon herring for prey.

In most biological systems, perfect synchrony among population components is rare. Instead, populations are often weakly synchronous to asynchronous and statistical

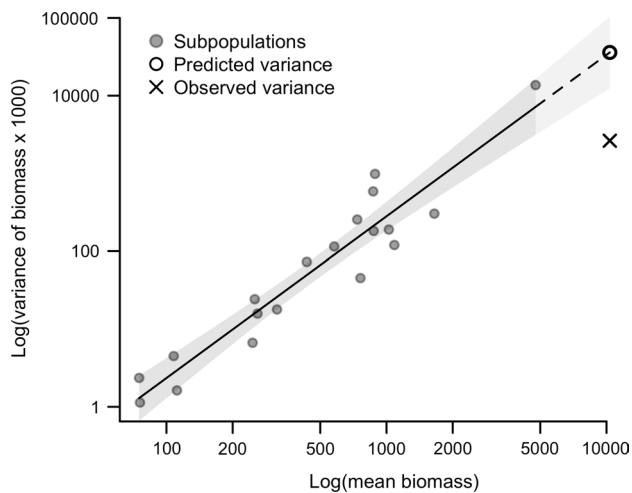


Fig. 5 Mean-variance relationship in Puget Sound herring population units (grey symbols). The open circle represents the variance expected for a population with the same adult spawning biomass as the aggregate Puget Sound population, based on a linear fit of the observed population mean and variance (solid line). The cross represents the observed variance for the aggregate Puget Sound population. Note the log scales on both axes

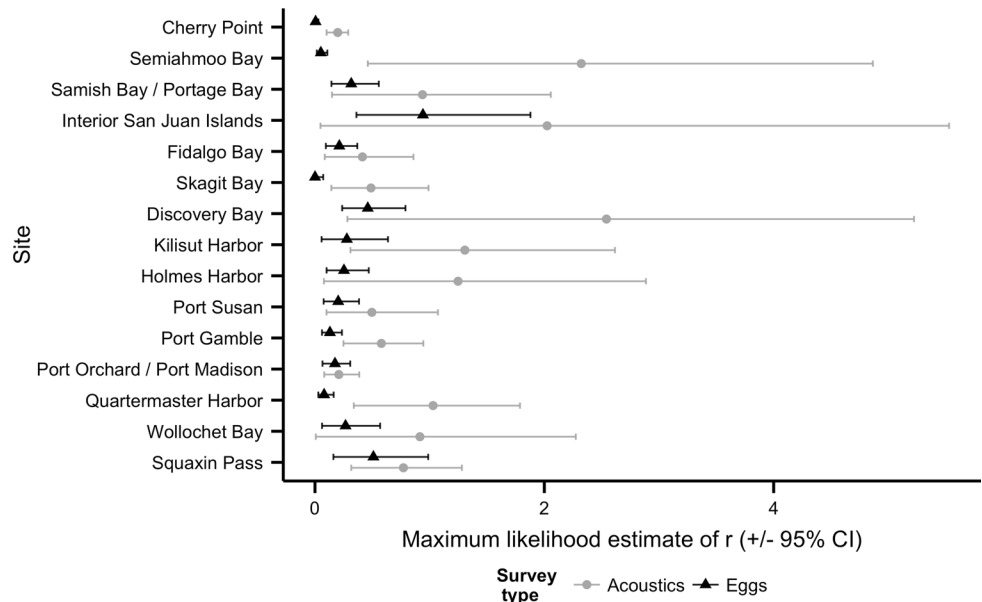
averaging causes aggregated populations to have a lower variance. Thus, PEs greater than zero are, to some extent, statistically inevitable. This statistical averaging has been used in the past to explain some of the diversity-stability relationships observed in nature (Doak et al. 1998). The properties of biological communities are also scale dependent; communities are known to be highly variable at some scales and less variable at others (Levin 1992). However, the combination of independent population dynamics and multiple spawning subpopulations overall contribute to a more stable herring population, which is of primary importance for a resource such as a forage fish that is heavily relied upon across the ecosystem.

Even though process variance was similar across stocks, the best-fit model here indicated that each egg survey time series measures a different subpopulation trajectory, suggesting that herring fluctuations vary at the local spawning site scale: good and bad years are not the same across subpopulations. The shared process variance for all subpopulations [$Q_{\text{diag}} = 0.09$ (0.04, 0.10)] and the absence of covariance ($Q_{\text{offdiag}} = 0$) in the best model show that the

Table 2 Model ranking and fits for hypotheses of different survey error structure

Hypothesis	ΔAIC_c	ΔAIC_b
Eggs and acoustics measure the same process, observation error varies by subpopulation	0	0
Eggs and acoustics measure the same process, observation error has same magnitude in each subpopulation	79.6	22.7
Eggs and acoustics measure different processes, observation error varies by subpopulation	42.8	144.7
Eggs and acoustics measure different processes, observation error has same magnitude in each subpopulation	121.0	626.1

Fig. 6 Mean observation errors $\pm 95\%$ bootstrapped CIs for acoustic (circles and grey lines) and egg (triangles and black lines) surveys at spawning sites where both survey methods were used. Dungeness Bay, Northwest San Juan Islands, Quilcene Bay, and South Hood Canal sites were removed because they had insufficient acoustic survey data



magnitude of variability in biomass is similar throughout the Puget Sound, but that there is very little covariance between these population units, confirming the independence indicated by the MARSS model-fitting exercise. The large magnitude of variance ($\mathbf{Q}_{\text{diag}}$) relative to the deterministic growth trend u [-0.01 ($-0.03, 0.01$)] suggests that variability is an important component of herring biomass dynamics, and this variability is probably the largest contributor to the observed dynamics. This matches what is observed in forage fish populations globally.

There are at least two ways to measure PEs from the biomass time series estimated by a MARSS model. Here, we measured PEs from the states estimated by the best-fit stock structure model. A second method involves estimating the full variance–covariance ($\mathbf{Q}$) matrix, whose elements describe variability at each site and covariance between sites (Table S3). Negative covariance in the off-diagonal elements of the $\mathbf{Q}$ matrix in this case would indicate asynchrony between the corresponding subpopulations, and 0 covariance elements would indicate independent dynamics. Any covariance less than unity between population components could lead to a PE. Using this method also resulted in portfolio-type effects, albeit with marginally weaker estimates of interannual variability ($\text{CV} = 0.28$ vs 0.30) and synchrony (synchrony index = 0.16 vs 0.23), with a wider confidence interval for the PE [mean–variance PE = 1.55 ($0.95, 2.53$)]. Regardless of method used, we found evidence of a PE in Puget Sound herring, and estimated values similar to, or larger than, PEs found in other animal populations [ranging from 0.5 to 2.0 in an empirical meta-analysis (Anderson et al. 2013)]. We also found that the effect was weaker [PE = 1.35 ($0.732, 2.509$)] for the latter 20 years of our data set than for the first 20 years [PE = 3.04 ($1.678, 5.508$)], and that the latter 20 years of the data set are also more synchronous (synchrony = 0.18 vs. 0.05). Although this difference is not statistically significant, it suggests that the stability of the Puget Sound herring population has changed over time.

The model selection procedure used here provides little conclusive evidence for PEs arising from consistent movements between populations, such as in a metapopulation, versus those arising from independence among subpopulations. Both PE methods and the best-fit MARSS model show little to no covariance among subpopulations in their biomass trends. Furthermore, we found little synchrony among subpopulations, and variable and inconsistent correlations between subpopulation pairs, all of which suggest independence among subpopulations as the source of the PE. However, it is possible for these same results to arise from movement of individuals between subpopulations, and previous work has shown that Pacific herring can change spawning beaches between years. For example, Pacific herring in British Columbia have site fidelity rates estimated between 75 and 96 % by studies of tagged adults

(Ware and Schweigert 2001). However, the tendency to return to a specific geographical area is scale dependent and Pacific herring in British Columbia returned more often to larger geographic areas [areas must be $\geq 500 \text{ km}^2$ to support high fidelity (Hay et al. 2001)]. It is possible that the population trends estimated by the MARSS model, and variation around those trends, are associated with movement by herring among different spawning areas from year to year. If spawners consistently move between neighboring beaches, then we would expect to see negative correlations between neighboring beaches (long-term negative correlations between neighboring beaches; Fig. 4) or negative covariance between beaches (year-to-year movements; Table S3). However, we found strong evidence for neither of these cases. While current genetic evidence shows that the bulk of the Puget Sound population units are more related to each other than to Squaxin Pass and Cherry Point, within that aggregate there is likely fine-scaled genetic structure that could identify patterns of homing, fidelity and movement. Future research to quantify the relatedness among Puget Sound herring population units will help elucidate the likely causes of observed population trends.

Stock identification is a top priority for fisheries and may benefit from the use of diverse methods. Herring stock structure is often defined on large geographical scales (e.g., across the British Columbia coast), or with the use of genetic and/or distance-based stock designations. Previous efforts to describe Puget Sound herring stock structure have hypothesized diversity in the stock based upon genetic data, spatial structure, and organic pollutants. Here, we were able to test how well these hypotheses describe biomass dynamics using MARSS models fitted to biomass data. Those tests returned the strongest support for the most fine-scale stock structure, i.e., 19 independent Puget Sound herring population units. Genetic, contaminant, and basin-wide models for population structure were poorer fits for biomass time series, suggesting that Puget Sound herring population units fluctuate at a finer scale than previously thought. We confirmed that our model selection process was capable of identifying population structure (shared dynamics between subpopulations) with time series data similar to those used here for Puget Sound herring using a simulation exercise (see Appendix S4).

This analysis suggests that at the scale of Puget Sound, herring biomass is relatively stable, but decreasing (Fig. 3). The 95 % confidence interval for the estimated trend spanned 0 [growth rate $u = -0.01$ ($-0.03, 0.01$)]. The weak trend likely results from the fact that population units are fluctuating independently overall and many show opposite trends. Indeed, some subpopulations—including those at Holmes Harbor, Quilcene Bay, Samish Bay/Portage Bay, and Squaxin Pass—show increasing trends, but the rest are declining. The Puget Sound-wide growth rate fitted here demonstrates the diversity in growth rates among stocks, while the variance–covariance matrix demonstrates

diversity in year-to-year variation. Observation error (R_{diag}) for surveys within Puget Sound is larger than process error (Q_{diag}), meaning that observed abundance patterns in the raw survey data are not the best descriptors of population size and variability. Although the overall biomass trend is not significantly declining, the detection of a PE indicates that the number of population units is important for stability, and that efforts to prevent local extinctions are likely important for maintaining this stability.

Differences between survey methods

One advantage of the state-space modeling approach is that multiple data sources can be used to estimate the same underlying demographic process. In doing so, we are also able to compare the accuracy of each monitoring tool in describing the underlying demographic processes, and potentially offer guidance for resource managers. In addition to providing the best support yet for Puget Sound herring population structure, the present analysis points to the use of estimated time series from an AR model as a more accurate reflection of underlying states than using simple long-term means or interpolation.

We found that both herring biomass survey methods estimated the same underlying process, i.e., a decreasing but non-significant trend in the aggregate herring population. However, the two methods varied in their error rates, with egg deposition surveys having higher accuracy ($r_{\text{egg avg.}} = 0.31$) than acoustic surveys on spawning aggregations ($r_{\text{acoustics avg.}} = 0.82$). Additionally, survey error rates varied by spawning site and method (Fig. 6). This suggests that although egg surveys are more reliable overall, there may be individual sites where one survey type is more important for capturing underlying biomass trends. For example, acoustic/trawl surveys are likely more accurate at sites with rocky substrate because rake surveys undersample spawn-on-gravel. Our results match observations by local biologists: in general, the spawning sites where our model struggled to estimate herring biomass using the egg survey data matched the set of sites identified by WDFW biologists as being less amenable to egg surveys; likewise, with the acoustic survey method and estimates (K. Stick, personal communication). At present, herring are surveyed using only the egg deposition surveys, which our model identified as being the more accurate survey method, although acoustic surveys may still be useful for gathering non-biomass data relevant to life history and stock assessment.

Environmental effects on herring biomass

This complex picture of Pacific herring populations—evidence for both a shared trend and independence—may be

the result of herring responding to their environment at multiple scales. The common growth suggests that herring respond synchronously to shared broad-scale drivers. Despite this result and previous results linking herring abundance to environmental drivers, however, we found that regional environmental covariates were only minimally important in describing shifts in herring adult spawner biomass. Previous research by Reum et al. (2011) showed that cumulative coastal upwelling positively influences the stock recruitment of Skagit Bay herring, but we found no Puget Sound-wide effect of upwelling. Herring recruitment in the Gulf of Alaska responds strongly to winter atmospheric pressure (Zebdi and Collie 1995) and we detected a minor effect of PDO on herring spawning biomass. Shifts in PDO have been associated with changes in zooplankton range and total biomass in the northeast Pacific, with years of negative PDO resulting in decreases in the biomass of zooplankton on the West Coast (Hare et al. 1999). The effect we estimated here [-0.0024 (-0.0042 , -0.0006)] may be weak because oceanic influences vary across space within Puget Sound, masking regional influences. For example, the Skagit River is closer to the Strait of Juan de Fuca, whereas herring spawning in central or southern Puget Sound may be less exposed to oceanic dynamics such as upwelling. It is likely that both local and regional environmental conditions interact to influence herring abundance, and future evaluations of environmental influences on herring should include subpopulations for which both recruitment and local and regional environmental data are available.

Although we find very limited model support for covariate effects, our analyses are not exhaustive and other potential influences on abundance time series should be considered. Potential influences on herring biomass not considered here include predation, disease, spawning ground habitat availability, and a zooplankton food source that may not be linearly related to any of the covariate data tested here. For example, Puget Sound harbor seal populations have increased in recent decades, and their intense predation on herring may reduce subpopulations size, regionally or locally. Loss of nearshore habitat associated with urbanization may also impact the population at local and regional scales. These other potential influences on herring warrant future exploration.

Our results demonstrate the importance of spatial scale and subpopulation diversity in the dynamics of Puget Sound herring, and that this diversity across a broad landscape confers stability upon the aggregate herring resource. The mechanisms behind these independent dynamics were not revealed by the present analysis, and warrant future research. Because of the critical role Pacific herring play in the Puget Sound food web, and the associated importance

of maintaining long-term stability in the population, resource managers and land-use planners would do well to consider the importance of maintaining a diversity of spawning habitats in the Puget Sound, rather than focusing on a few highly productive sites.

Acknowledgments The authors would like to thank E. E. Holmes, M. D. Scheuerell, E. J. Ward, D. E. Schindler, A. O. Shelton and two anonymous reviewers for reviewing earlier versions of this manuscript. We thank Kurt Stick, Adam Lindquist and Dayv Lowry at the Washington Department of Fish and Wildlife for data and for their insight on Puget Sound Pacific herring, and Kiva Oken for assistance with simulating correlated time series. M. C. Siple was supported by an NSF Graduate Research Fellowship and a fellowship from the University of Washington School of Aquatic and Fishery Sciences. This document has been funded wholly or in part by the United States Environmental Protection Agency under assistance agreement 00J30301-0 to the University of Washington. The contents of this document do not necessarily reflect the views and policies of the Environmental Protection Agency, nor does mention of trade names or commercial products constitute endorsement or recommendation for use.

Author contribution statement T. B. F. conceived the study, M. C. S. designed model selection procedures and performed the analysis. M. C. S. and T. B. F. wrote the manuscript.

Compliance with ethical standards

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare that they have no conflict of interest.

References

- Agashe D (2009) The stabilizing effect of intraspecific genetic variation on population dynamics in novel and ancestral habitats. *Am Nat* 174:255–267. doi:10.1086/600085
- Anderson EM, Lovvorn JR, Esler D et al (2009) Using predator distributions, diet, and condition to evaluate seasonal foraging sites: sea ducks and herring spawn. *Mar Ecol Prog Ser* 386:287–302. doi:10.3354/meps08048
- Anderson SC, Cooper AB, Dulvy NK (2013) Ecological prophets: quantifying metapopulation portfolio effects. *Methods Ecol Evol* 4:971–981. doi:10.1111/2041-210X.12093
- Beaudreau AH, Essington TE (2011) Use of pelagic prey subsidies by demersal predators in rocky reefs: insight from movement patterns of lingcod. *Mar Biol* 158:471–483. doi:10.1007/s00227-010-1574-6
- Burton SF (1990) Comparison of Pacific spawner herring biomass estimates from hydroacoustic-trawl and spawning ground escapement surveys in Puget Sound, Washington
- Casini M, Kornilovs G, Cardinale M et al (2011) Spatial and temporal density dependence regulates the condition of central Baltic Sea clupeids: compelling evidence using an extensive international acoustic survey. *Popul Ecol* 53:511–523. doi:10.1007/s10144-011-0269-2
- Cavanaugh JE, Shumway RH (1997) A bootstrap variant of AIC for state-space model selection. *Stat Sin* 7:473–496
- Chapman WM, Katz M, Erickson DW et al (1941) The races of herring in the state of Washington
- Doak DF, Bigger D, Harding EK et al (1998) The statistical inevitability of stability-diversity relationships in community ecology. *Am Nat* 151:264–276. doi:10.1086/286117
- Duffy EJ, Beauchamp DA, Sweeting RM et al (2010) Ontogenetic diet shifts of juvenile chinook salmon in nearshore and offshore habitats of Puget Sound. *Trans Am Fish Soc* 139:803–823. doi:10.1577/t08-244.1
- Grabowski TB, Thorsteinsson V, McAdam BJ, Marteinsdóttir G (2011) Evidence of segregated spawning in a single marine fish stock: sympatric divergence of ecotypes in Icelandic cod? *PLoS One* 6:e17528. doi:10.1371/journal.pone.0017528
- Hanski I (1999) *Metapopulation ecology*. Oxford University Press, Oxford, New York
- Hare SR, Mantua NJ, Francis RC (1999) Inverse production regimes: Alaska and West Coast Pacific Salmon. *Fisheries* 24:6–14. doi:10.1577/1548-8446
- Hatch S (2013) Kittiwake diets and chick production signal a 2008 regime shift in the Northeast Pacific. *Mar Ecol Prog Ser* 477:271–284. doi:10.3354/meps10161
- Hay DE, McCarter PB, Daniel KS (2001) Tagging of Pacific herring *Clupea pallasii* from 1936 to 1992: a review with comments on homing, geographic fidelity, and straying. *Can J Fish Aquat Sci* 58:1356–1370. doi:10.1139/cjfas-58-7-1356
- Hinrichsen R, Holmes E (2009) Using multivariate state-space models to study spatial structure and dynamics. In: Cantrell R, Cosner C, Ruan S (eds) *Spatial ecology*. CRC/Chapman Hall, Boca Raton, pp 145–166
- Holmes EE (2014) Derivation of an EM algorithm for constrained and unconstrained multivariate autoregressive state-space (MARSS) models. arXiv preprint arXiv:1302.3919
- Holmes EE, Ward EJ, Wills K (2012) Marss: multivariate autoregressive state-space models for analyzing time-series data. *R J* 4:11–19
- Hooper DU, Chapin FS, Ewel JJ et al (2005) Effects of biodiversity on ecosystem functioning: a consensus of current knowledge. *Ecol Monogr* 75:3–35. doi:10.1890/04-0922
- Hutchinson WF (2008) The dangers of ignoring stock complexity in fishery management: the case of the North Sea cod. *Biol Lett* 4:693–695. doi:10.1098/rsbl.2008.0443
- Hutchinson WF, Carvalho GR, Rogers SI (2001) Marked genetic structuring in localised spawning populations of cod *Gadus morhua* in the North Sea and adjoining waters, as revealed by microsatellites. *Mar Ecol Prog Ser* 223:251–260. doi:10.3354/meps223251
- Ives AR, Dennis B, Cottingham KL, Carpenter SR (2003) Estimating community stability and ecological interactions from time series data. *Ecol Monogr* 73:301–330. doi:10.1890/0012-9615(2003)073[0301:ECSAEI]2.0.CO;2
- Lance MM, Jeffries SJ (2006) Estimating importance of rockfish, lingcod and other bottomfish in the diet of harbor seals in the San Juan Islands. Washington Department of Fish and Wildlife, Olympia
- Lehman CL, Tilman D (2000) Biodiversity, stability, and productivity in competitive communities. *Am Nat* 156:534–552. doi:10.1086/303402
- Levin SA (1992) The problem of pattern and scale in ecology. *Ecology* 73:1943–1967. doi:10.2307/1941447
- Lindegren M, Andersen KH, Casini M, Neuenfeldt S (2014) A meta-community perspective on source-sink dynamics and management: the Baltic Sea as a case study. *Ecol Appl* 24:1820–1832. doi:10.1890/13-0566.1
- Loreau M, de Mazancourt C (2008) Species synchrony and its drivers: neutral and nonneutral community dynamics in fluctuating environments. *Am Nat* 172:E48–E66. doi:10.1086/589746
- MacArthur R (1955) Fluctuations of animal populations and a measure of community stability. *Ecology* 36:533–536. doi:10.2307/1929601

- McArdle BH, Gaston KJ, Lawton JH (1990) Variations in the size of animal populations—patterns, problems and artifacts. *J Anim Ecol* 59:439–454. doi:[10.2307/4873](https://doi.org/10.2307/4873)
- McQuinn IH (1997) Metapopulations and the Atlantic herring. *Rev Fish Biol Fish* 7:297–329. doi:[10.1023/a:1018491828875](https://doi.org/10.1023/a:1018491828875)
- Michener WK, Baerwald TJ, Firth P et al (2001) Defining and unraveling biocomplexity. *Bioscience* 51:1018–1023. doi:[10.1641/0006-3568\(2001\)051\[1018:DAUB\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2001)051[1018:DAUB]2.0.CO;2)
- National Marine Fisheries Service Fisheries Statistics and Economics Division (2013) Current Fisheries Statistics No. 2013
- Penttilä D (2007) Marine forage fishes in Puget Sound. Washington Department of Fish and Wildlife
- Reum JCP, Essington TE, Greene CM et al (2011) Multiscale influence of climate on estuarine populations of forage fish: the role of coastal upwelling, freshwater flow and temperature. *Mar Ecol Prog Ser* 425:203–215. doi:[10.3354/meps08997](https://doi.org/10.3354/meps08997)
- Schindler DE, Hilborn R, Chasco B et al (2010) Population diversity and the portfolio effect in an exploited species. *Nature* 465:609–612. doi:[10.1038/nature09060](https://doi.org/10.1038/nature09060)
- Schindler DE, Armstrong JB, Bentley KT et al (2013) Riding the crimson tide: mobile terrestrial consumers track phenological variation in spawning of an anadromous fish. *Biol Lett* 9:20130048. doi:[10.1098/rsbl.2013.0048](https://doi.org/10.1098/rsbl.2013.0048)
- Schindler DE, Armstrong JB, Reed TE (2015) The portfolio concept in ecology and evolution. *Front Ecol Environ* 13:257–263. doi:[10.1890/140275](https://doi.org/10.1890/140275)
- Schrimpf MB, Parrish JK, Pearson SF (2012) Trade-offs in prey quality and quantity revealed through the behavioral compensation of breeding seabirds. *Mar Ecol Prog Ser* 460:247–259. doi:[10.3354/meps09750](https://doi.org/10.3354/meps09750)
- Schweigert J, Haegele C (2001) Estimates of egg loss in Pacific herring spawning beds and its impact on stock assessments. University of Alaska Sea Grant College Program report herring: expectations for a new millennium AK-SG-01-04:489–508
- Shelton AO, Francis TB, Williams GD et al (2014) Habitat limitation and spatial variation in Pacific herring egg survival. *Mar Ecol Prog Ser* 514:231–245. doi:[10.3354/meps10941](https://doi.org/10.3354/meps10941)
- Small MP, Loxterman JL, Frye AE et al (2005) Temporal and spatial genetic structure among some Pacific herring populations in Puget Sound and the southern Strait of Georgia. *Trans Am Fish Soc* 134:1329–1341. doi:[10.1577/t05-050.1](https://doi.org/10.1577/t05-050.1)
- Stick K, Lindquist A, Lowry D (2014) 2012 Washington State herring stock status report
- Stout HA, Gustafson RG, Lenarz WH et al (2001) Status review of Pacific herring (*Clupea pallasii*) in Puget Sound, Washington
- Thorson JT, Scheuerell MD, Buhle ER, Copeland T (2014) Spatial variation buffers temporal fluctuations in early juvenile survival for an endangered Pacific salmon. *J Anim Ecol* 83:157–167. doi:[10.1111/1365-2656.12117](https://doi.org/10.1111/1365-2656.12117)
- Trites AW, Calkins DG, Winship AJ (2007) Diets of Steller sea lions (*Eumetopias jubatus*) in Southeast Alaska, 1993–1999. *Fishery Bulletin* 105:234–248
- Ward EJ, Chirakkal H, Gonzalez-Suarez M et al (2010) Inferring spatial structure from time-series data: using multivariate state-space models to detect metapopulation structure of California sea lions in the Gulf of California, Mexico. *J Appl Ecol* 47:47–56. doi:[10.1111/j.1365-2664.2009.01745.x](https://doi.org/10.1111/j.1365-2664.2009.01745.x)
- Ware D, Schweigert J (2001) Metapopulation structure and dynamics of British Columbia herring. Department of Fisheries and Oceans, Ottawa
- West JE, O'Neill SM, Ylitalo GM (2008) Spatial extent, magnitude, and patterns of persistent organochlorine pollutants in Pacific herring (*Clupea pallasii*) populations in the Puget Sound (USA) and Strait of Georgia (Canada). *Sci Total Environ* 394:369–378. doi:[10.1016/j.scitotenv.2007.12.027](https://doi.org/10.1016/j.scitotenv.2007.12.027)
- Zebdi A, Collie JS (1995) Effect of climate on herring (*Clupea pallasii*) population dynamics in the Northeast Pacific Ocean. *Can Spec Publ Fish Aquat Sci* 121:277–290